

prey See *predator*.

prime number Any positive whole number that cannot be divided to give another whole number except by dividing by itself and 1.

prism (1) Any solid geometric form with sides that form a parallelogram. (2) A prism-shaped block of glass, especially one with triangular sides, that is used to split white light into the colors of the spectrum.

probability The likelihood of an event happening, normally expressed as a value between 0 and 1.

product In mathematics, the result of one number being multiplied by another.

prokaryotic cell The cell of microscopic organisms such as bacteria: it is smaller than a *eukaryotic cell* and does not have a separate nucleus.

prosthetic An artificial body part.

protein Any of thousands of different types of large molecules made by the body and coded for by genes. See also *amino acid*.

proton A particle in the nucleus of an atom that has a positive electric charge.

pulsar A neutron star which can be detected because its rapid rotation causes pulses of radiation to be beamed outward.

quadrant A navigational instrument.

quadratic equation Mathematical equation containing at least one variable number that is multiplied by itself once (for example xxx , also written x^3), but none which have been multiplied more times than this.

quantum electrodynamics The quantum physics theory that deals with the interactions between electrons, positrons, and photons.

quantum physics The branch of science that deals with subatomic particles and energy interactions in terms of minute discrete energy packets called quanta.

quantum theory The theory that light and other electromagnetic radiation is made up of a stream of photons, each carrying a certain amount of energy.

quark One of a group of fundamental particles that do not exist separately but make up protons, neutrons, and some other subatomic particles.

quasar An immensely powerful source of radiation occurring beyond our own galaxy. Quasars are thought to be the central regions of other galaxies that are producing far more radiation than the Milky Way's center does.

radar A way of detecting objects by sending out radio waves and collecting their returning "echoes."

radiation Any stream of fast-moving particles or waves.

radio waves Invisible waves at the low-frequency end of the electromagnetic spectrum, whose wavelength can range from kilometers to centimeters (microwaves).

radioactive Emitting high-energy subatomic particles or radiation as part of radioactive decay.

radioactive decay The process whereby unstable nuclei emit high-energy particles, or radiation, as they break up or transform.

radioactive tracers Substances that contain radioactive atoms to allow easier detection and measurement.

radioactivity Phenomena involving radioactive decay.

radiometric dating The process of finding a rock's absolute age by detecting the stage of radioactive decay of particular isotopes in it.

RAM Random access memory. Computer memory chips where information can be stored and retrieved.

rarefaction The opposite of compression, especially in a gas, where the density becomes lowered.

ratio The proportional relation between two numbers.

reactant A substance taking part in a chemical reaction.

reaction (1) A force same in magnitude, but opposite in direction, to another force. Every force has a reaction. (2) Any change that alters the chemical properties of a substance or forms a new substance.

red giant A star that is nearing the end of its life and has expanded to a giant size and become reddish in color.

red shift The tendency for wavelengths of light to be shifted toward the red end of the spectrum when their source is receding rapidly from the observer. The effect also occurs with other wavelengths of electromagnetic radiation.

reduction Originally, a reaction where a substance loses oxygen; now used more generally for any reaction where a substance gains electrons. Its opposite is oxidation.

reflex An automatic reaction to something.

refraction The bending of light rays as they enter a different medium at an angle, such as from air to water.

refractive index The ratio of the speed of light in one medium to the speed of light in a second medium.

relativity The description of space and time, energy and matter according to the theories of Albert Einstein, which depend on the constancy of the speed of light in a vacuum.

repoussé The ancient art of decorating metal by hammering on the back of the piece.

reproduction Process of creating offspring.

resistance See *electrical resistance*.

resistor An electrical device or component that resists current flow.

resonance Situation when the vibrations of an object become large because it is being made to vibrate at its "natural" frequency.

respiration (1) Breathing. (2) Also called cellular respiration, the biochemical processes within cells that break down food molecules to provide energy, usually by combining the food molecules with oxygen.

retrograde motion Motion that is the opposite of another motion, such as a satellite orbiting a planet in the opposite direction to the planet's own rotation. Retrograde motion can only be apparent as when a planet seems to move backward against the stars because Earth is overtaking it on its orbit round the Sun.

retrovirus An RNA virus—such as HIV—which reproduces itself by inserting a DNA copy of its genes into the host cell.

Richter scale A scale for measuring the size of an earthquake in terms of the amount of energy released.

right angle An angle created by lines meeting perpendicularly, forming an equal angle on either side.

RNA Ribonucleic acid. A *molecule* similar to DNA with various roles in cells, including acting as an intermediary between DNA and the rest of the cell.

robot (1) An intelligent humanlike machine (mainly in fictional contexts). (2) A machine, especially a programmable one, that can carry out a complex series of movements.

ruminant An animal such as a cow or deer that chews the cud.

S wave Secondary wave. An earthquake wave that travels through the ground as lateral or horizontal waves.

satellite An object that orbits a planet. There are natural satellites, such as a moon, and artificial satellites, such as a craft used to retransmit radio signals.

secretion The release of specific substances by the cells of living things.

sedimentary rock Formed when fragments of material settle on the floor of a sea or lake and are cemented together over time.

sedimentation The geological process in which loose material is laid down on ocean and river beds, and by wind and moving ice.

seismic wave A wave that travels through the ground, such as from an earthquake.

seismograph A device for measuring and recording earthquake waves.

seismology The study of earthquakes.

seismometer A device for measuring earthquake waves. The term is now interchangeable with *seismograph*, since modern seismometers record their measurements as well.

selective breeding The process of choosing particular domestic animals for breeding to encourage the development of desired traits over the generations.

semiconductor A substance whose resistance is intermediate between a conductor and an insulator. The properties of semiconductor devices can be controlled and altered very exactly, making them vital in modern electronics.

sensory nerve A nerve that transmits information about the environment (touch, taste, etc.) to the central nervous system.

sepal One of a set of petal-like or leaflike structures usually found around the rim or base of a flower outside the petals.

sex cell See *gamete*.

sextant A navigational instrument designed to measure the altitude of an object, such as the Sun above the horizon at noon.

sexual reproduction Reproduction that involves the fusion of two gametes (sex cells) to produce a new individual.

SI unit A unit in the international system of measure based on the meter, kilogram, second, ampere, kelvin, candela, and mole.

silicon A semi-metallic element related to carbon, which is a constituent of many of Earth's rocks.

sine (sin) The ratio of the opposite side of an angle to the hypotenuse in a right-angled triangle. Also, the mathematical function describing how this ratio varies when the angle changes as the hypotenuse sweeps round a circle.

skeleton The frame of bone and cartilage in vertebrates that supports the body and protects its organs, or any structure in other animals that serves similar functions.

slide rule A ruler with a central sliding bar designed to make quick calculations using logarithms.

smelting Extracting a metal from its ore.

software The programs used by a computer.

solar constant The amount of heat energy from the Sun received per unit area of Earth's surface.

solar eclipse See *eclipse*.

solar flare A sudden burst of radiation from the Sun.

Solar System The system consisting of the Sun, the planets, and other objects orbiting the Sun, and the surrounding regions of space in which the Sun's influence is discernable.

solenoid A cylindrical coil of wire that becomes a magnet when an electric current is passed through it.

solstice Each of the two times in the year, one at midsummer and one at midwinter, when the Sun reaches its highest or lowest point in the sky at noon.

solubility The ability of a *solute* to dissolve.

solute The substance that dissolves in a solvent to form a solution.

solution A liquid in which individual atoms, molecules, or ions of another substance (as distinct from small solid particles) are evenly dispersed.

solvent A substance, especially a liquid, that can dissolve other substances.

somatic nuclear transfer A laboratory technique for creating a fertilized ovum using a somatic cell (ordinary body cell) to create a clone of the organism.

sonar A means of detecting objects and navigating under water by sending out sound waves and receiving their echoes.

space probe An unpowered vehicle (other than an Earth satellite) that is designed to explore space.

space station A human-occupied orbiting structure for the purposes of carrying out experiments, observations, etc.

space-time The three dimensions of space combined with time in a single continuum.

species A particular kind of living thing, often defined in terms of the ability of individuals of that species to mate and produce fully fertile offspring (although this definition does not work for all cases).

specific heat capacity The quantity of heat required to raise the temperature of a unit mass of a given substance by one degree.

spectroscope A machine that measures and analyses *spectra*.

spectroscopy The study and measurement of *spectra*.

spectrum (plural: **spectra**) Originally, light separated by refraction so that its different